

# A Nobel Nose: The 2004 Nobel Prize in Physiology and Medicine

## Essay

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On December 10, 2004, the anniversary of industrialist and philanthropist Alfred Nobel's death, the eponymous prize in Physiology or Medicine bequeathed by his prescient will was awarded to Linda Buck and Richard Axel for their seminal work in understanding olfaction. It is the first Nobel Prize in Physiology or Medicine for work reported in the decade of the 1990s, a relatively short time ago by the Nobel metric. Of particular significance, Linda Buck is only the seventh woman in its history to receive the prize in this category.

As it happens, this is not the first time the olfactory system has figured in a Nobel Prize. Nearly a century earlier, in 1906, this same prize was awarded to Ramon y Cajal for his formulation of the Neuron Doctrine, which, as he pointed out in his Nobel lecture (available, as are many other resources referred to in this article, at [Nobelprize.org](http://Nobelprize.org)), was based on work in the cerebellum, the spinal cord, the retina, and *the olfactory bulb*. Now, 98 years later, the olfactory system again has provided the Nobel committee with the opportunity to recognize pioneering achievements that will impact on our wider understanding of the brain.

According to the Nobel Prize press release, this year's prize was awarded for discoveries of "odorant receptors and the organization of the olfactory system." It recognizes not only the seminal paper appearing in the April 5, 1991, edition of *Cell*, "A Novel Multigene Family May Encode Odorant Receptors: A Molecular Basis for Odorant Recognition" (Buck and Axel, 1991), but also a considerable body of subsequent work by them, pursued independently and in parallel, which has further illuminated the logic of what had been the most enigmatic of our sensory systems.

Situated more or less in the center of your face, your nose is arguably the best chemical detector on the planet. Certainly for most animals it is their primary source of information about the world and determines what they will eat, when they should run away, and who will mate with whom. The system is thus a key arbiter in both survival and fitness. Even in humans there is a very well developed sense of smell, but the greatest obstacle to better functioning is our bipedalism: our noses tend to be a couple of meters in the air, while most odors are within 10–20 cm of the ground.

The peripheral olfactory system of mammals is able to detect many thousands of mostly organic low-molecular weight compounds with various functional groups, including aldehydes, alcohols, esters, ketones, amines, thiols, etc. Beginning at the periphery, but probably in central structures as well, the olfactory sys-

tem makes precise discriminations not only among these functional groups but between molecules that may differ by only a single carbon atom, as for example between the six and seven carbon aliphatic aldehydes that smell to us, respectively, like grass and soap. There are numerous cases of perceptual differences even between enantiomeric pairs of atomically identical compounds.

Although there appears to be a bewildering number and variety of odorous stimuli, it was imagined, in analogy with the visual system, that a handful of "primary" receptors acting with overlapping sensitivities could account for the wide sensitivity and discriminatory abilities of the system (Amoore, 1967). It was the acumen of Buck and Axel, and the data from their landmark paper, that unmasked the previously unimagined large family of receptors that mediate peripheral olfaction. Indeed the >1000 receptor genes they found are now recognized in the post-genome world as constituting the largest gene family in the mammalian genome and, as the total number of our genes seems to continue downward, represents an increasingly significant proportion of the genes in the entire genome (1%–4%).

How, precisely, did Buck and Axel come upon this genetic treasure chest? Through the 1980s, the olfactory system had been attracting new investigators among biochemists and molecular biologists, joining the ranks of physiologists and anatomists who had been studying the system for some years. Together these several groups provided new insights into olfactory transduction. Not unimportantly, this was around the same time that advances were being made in solving phototransduction with its second messenger cascade based on cGMP and a new kind of ion channel directly gated by intracellular cGMP (Schwartz, 1985; Yau, 1994). Using the then new patch-clamp technique to record odor-induced responses in olfactory neurons, an inward positive current was shown to underlie the odor-induced depolarization recorded many years earlier (Firestein and Werblin, 1989; Kurahashi, 1989; Lettvin and Gesteland, 1965). The kinetics of this response suggested the participation of a second messenger system. Biochemical work from Doron Lancet's laboratory identified cAMP as the likely second messenger (Pace et al., 1985), and work in Sol Snyder's lab showed an odor-induced, GTP-dependent rise in adenylyl cyclase activity (Sklar et al., 1986), providing the first suggestion that odorant detection was perhaps mediated by a G protein-coupled receptor (GPCR). Shortly thereafter, Randy Reed cloned from olfactory sensory neurons the gene for a new G protein subunit,  $G_{\alpha}$ -olf, further substantiating this view (Jones and Reed, 1989). Further evidence for a cAMP-based second messenger system subsequent to activation of an odorant-sensitive GPCR accumulated rapidly from biochemical, molecular, and physiological data (Boekhoff et al., 1990; Firestein et al., 1991; Reed, 1990). Perhaps the most important of those was the discovery, by the late Geoffrey Gold (Nakamura and Gold, 1987), of an ion channel sensitive to cAMP (and cGMP, although that

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is not the olfactory second messenger) on the cilia of olfactory neurons—odor sensitive structures functionally not unlike the outer segments of retinal rods and cones. Several laboratories cloned the genes for these channels (Cook et al., 1987; Dhallan et al., 1990; Ludwig et al., 1990), which turned out to be very closely related to the CNG channels first identified in rods. Unintuitively then, olfactory and photo transduction appeared quite similar. Lancet in an influential review article (Lancet, 1986) suggested that indeed all olfactory neurons shared a common transduction pathway and differed mainly in the receptor they expressed. That this receptor was likely to couple through a G protein, as did rhodopsin, was strongly suspected, although there was no direct evidence and there were several competing theories.

Into this setting came Linda Buck, then a senior post doc in Richard Axel's lab. The story of their key discovery has recently been recounted by Buck in a short review for *Cell's* 30th anniversary issue (Buck, 2004). It is worthwhile reading for any young scientist; and older ones as well. Buck and Axel embarked on the near quixotic search for mammalian odorant receptors with three key assumptions: that the receptors would be structurally related to the other 20 or so GPCRs then known, that they would be encoded by a large multigene family (although even they could not have imagined how large), and that expression of these genes would be restricted to the olfactory epithelium.

It was here that Buck and Axel brought together their modest assumptions and the still relatively new technique of PCR in an elegant and ultimately successful experimental program. They first designed 11 degenerate PCR primers that matched conserved regions of known GPCRs (assumption 1). They then devised a novel combinatorial PCR strategy in which they used these primers in all 30 possible pairwise combinations in PCR reactions with cDNA prepared from rat olfactory epithelium (assumption 3). From this they obtained over 60 PCR products in the appropriate size range. But short of a massive sequencing effort, not as easily available in the late 1980s as it is now, how could they identify which, if any, of these products encoded odorant receptors? In another novel approach, they reasoned that if their degenerate primers were hybridizing with numerous members of the presumptive olfactory gene family, then a band on the agarose gel actually represented many different, although similar, genes (assumption 2). If so, restriction enzymes that cut frequently would cut each amplified gene segment differently and would thus produce different, multiple sets of fragments that would run at different weights in a gel. Therefore, a sample that was made up of many different genes would, upon digestion, produce a host of bands, while a sample consisting of a single gene would produce relatively fewer bands (because all the cuts would be in the same place). In fact, they reasoned, the molecular weights of the fragments from a family of genes would add up to much more than the molecular weight of the original sample, a deceptively simple assay for the existence of multiple related genes such as might be expected in a large family of GPCRs. Although not often appreciated, the combination of multiple degenerate primers and frequently cutting re-

striction enzymes was one of the earliest, and certainly the most sophisticated, uses of PCR as an experimental technique, rather than simply as a biochemical tool for producing DNA.

In their classic paper, Buck and Axel show one set of agarose gels with the now famous "lane 13" showing just such a pattern of multiple bands. They had indeed found the first odorant receptors (ORs). This original paper is still worth reading, not only for the elegance of the experimental approach, but for the remarkable prescience of the authors in their interpretation of the data. They recognized that the variability between the related odorant receptors was especially concentrated in regions recently predicted to be a putative binding site for other GPCR ligands (Strader et al., 1989), and this immediately suggested a molecular basis for odor recognition. They were able to show that the odorant receptor family had at least 100 to 200 members, but predicted that it could be far larger—even the largest family in the genome. They recognized that odorant receptors, in common with many other GPCRs, were encoded by a single exon and that this meant that gene rearrangement or alternative splicing were not likely mechanisms for generating diversity in the receptor family—rather, each gene encoded a specific single receptor. They rightly intuited, from the frequency of specific OR sequences in a cDNA library, that any given sensory neuron was likely to express only one or a very few of the receptor genes. Many of these predictions have been borne out by experimental activity spurred on by this paper and continuing for the last 14 years.

All at once, immense new possibilities opened up in the field of olfaction. Ideas for diverse types of coding mechanisms that had been bandied about for some time were suddenly either possible or easily dismissed. For example, the possibility that, by analogy to color vision, there might be only a handful of receptors tuned to "primary odorants" was now excluded (there were hundreds of odorant receptors, while there were only three "color receptors"), as were the variety of corollary chemical schemes depending on primary odors. Indeed, with this paper it now became clear that the olfactory system was uniquely suited to investigation by molecular techniques. Over the next decade, Buck and Axel and their colleagues, working independently at Harvard and Columbia, respectively, used odorant receptor genes to tackle the organization of the olfactory system.

In the olfactory epithelium, Buck and Axel found that each receptor gene is expressed in only a small percentage of neurons, which are randomly distributed within one of several expression zones (Ressler et al., 1994; Vassar et al., 1994). Although the role of this zonally restricted expression remains unknown, it strongly mitigates against a complex epithelial odor map. It further suggested that each neuron might express only one receptor gene, a scenario in which signals from different receptors would be segregated in different neurons. The expression of one receptor gene per neuron was later confirmed by Buck's group using single-cell RT-PCR (Malnic et al., 1999). Together with Axel's finding that the two alleles of a particular receptor gene are expressed by different subpopulations of neurons (Chess et al., 1994), this established that receptor ex-

pression is both monogenic and monoallelic (Mombaerts, 2004).

Subsequently, Buck's laboratory undertook a detailed investigation of the molecular basis for a combinatorial code in the peripheral olfactory system. Using calcium imaging and single-cell RT-PCR in a kind of technical tour de force, Malnic et al. (1999) showed how subtle changes in ligand (odorant) structure alters the number and identity of receptors activated. Thus, any given odorant receptor may recognize a variety of related odor ligands, and a particular odor ligand can bind to numerous receptors, creating a nearly limitless matrix of combinations that allows for unequaled discrimination of the complex chemical world.

Although its initial impact was to establish a molecular framework for odorant recognition, the original discovery of the receptors has provided insights into a string of systems-level questions. How could the brain make sense of input from so many populations of cells expressing different receptors? There must, after all, be at least 1000 "labeled lines," that is, populations of several thousand sensory neurons expressing a particular receptor. Further, the axons of some 15,000 olfactory sensory neurons converge onto fewer than 25 of the second-order mitral cells in the brain (Treloar et al., 2002). How was this organized for the brain to extract useful information from the primary sensory inputs? To approach these questions, the Buck and Axel groups turned their attention to the first relay in the olfactory system, the olfactory bulb. Here the axons of epithelial sensory neurons synapse onto mitral cell dendrites in structures called "glomeruli," of which there are about 2000 per bulb in rat and mouse. Relying on the possible presence of OR mRNA in the axon terminals of olfactory sensory neurons, Buck and Axel used *in situ* hybridization and found that different receptor probes labeled a small set of different glomeruli (Ressler et al., 1994; Vassar et al., 1994). These data were soon extended through the application of an elegant gene-targeting approach developed by Mombaerts, initially as a post doc in Axel's laboratory and subsequently in his own laboratory. By targeting markers such as *lac-z* or GFP into a specific OR locus, thus marking all cells expressing a particular receptor—and none others—they were able to overcome the numerical challenge presented by the large number of ORs and gain a precise view of how neurons expressing particular receptors were mapped onto the olfactory bulb (Mombaerts et al., 1996a). These studies produced remarkably clear results, as well as strikingly beautiful photographs, showing that olfactory sensory neurons expressing the same receptors were distributed over large areas of epithelium but that all of their axons converged to a few specific glomeruli at more-or-less stereotyped sites in the bulb. They further established that each glomerulus is likely dedicated to one receptor and that, as in the epithelium, signals from different receptors are segregated.

In subsequent experiments, the Buck lab explored how information from different odorant receptors is organized in the olfactory cortex, the next relay in the olfactory system. Using a novel genetic tracing system (Horowitz et al., 1999), they prepared mice that coexpressed a transneuronal tracer with a single receptor

gene (Zou et al., 2001). The tracer traveled from neurons expressing that particular receptor to recipient glomeruli, to projection neurons in the bulb, and then to cortical neurons postsynaptic to the bulb neurons. The tracer labeled five to six distinct clusters of neurons in the olfactory cortex, the locations of which were similar among individuals, but different when coexpressed with different receptor genes, providing a first glimpse of signal organization in the olfactory cortex.

Taken together, these studies represent the first instance in which the wiring diagram of a neural circuit has been revealed by molecular biology. Decades of electrophysiological studies provided information about responses to odorants by neurons in the epithelium and in glomeruli and projection neurons in the bulb (reviewed in Shepherd et al., 1975; Kauer, 1987; Mori et al., 1999). However, it was with the molecular analysis of odorant receptor expression and the neural pathways followed by individual receptor inputs that the logic underlying the system came into focus. With the discoveries by Buck and Axel came a host of new questions that captured the imagination of new investigators as well as old. Of particular interest to many have been the developmental mechanisms by which each olfactory sensory neuron selects a single receptor gene for expression and those underlying the convergence of like axons in specific glomeruli in the bulb. Remarkably, the receptor expressed by a neuron is a critical, and perhaps the sole, determining factor in the targeting of a sensory axon to a particular glomerulus in the bulb (Mombaerts et al., 1996b; Wang et al., 1998; Feinstein et al., 2004). Changing the receptor expressed by a sensory neuron alters its glomerular projection. Thus, the receptor itself imparts the identity of the olfactory neuron—not only regarding what odors will stimulate it, but also where in the brain it will project (Feinstein and Mombaerts, 2004). This is an entirely new function for GPCRs, and one wonders whether this may be more widespread among other neurons expressing GPCRs for other transmitters in the nervous system.

Also critical to the development of this story was the dawn of whole-genome genomics. A family of more than 1000 genes could not be handled easily with standard molecular techniques. However, they are ideally suited to bioinformatic and computational analysis. With the availability of the sequenced human genome, Lancet and others (Glusman et al., 2001; Zozulya et al., 2001) identified some 350 human ORs with intact open reading frames—a family smaller than in rat or mouse but still of considerable size (for comparison, the next-largest family of nonchemosensory GPCRs is the serotonin receptors—consisting of a whopping 15 genes!). Interestingly, an additional 600 or so OR sequences were discovered in the human genome, but they were all pseudogenes, containing one or more sequence disruptions that would render the resulting protein nonfunctional. A year later, the mouse genome was available, and several data mining efforts discovered more than 1200 OR genes, including 250 or so pseudogenes (Lane et al., 2001; Zhang and Firestein, 2002). High fractions of pseudogenes are seen in all animals analyzed so far, suggesting a dynamically evolving gene family.

The discovery of the mammalian ORs informed and



motivated the search for other families of chemosensory receptors. They have now been followed by the discovery of two large families of receptors (about 150 to 200 genes each) in the vomeronasal organ of rat and mouse (Dulac and Axel, 1995; Matsunami and Buck, 1997; Rodriguez et al., 2002; Herrada and Dulac, 1997; Ryba and Tirindelli 1997), believed to be specialized for pheromone detection; a smaller but significant family of about 100 receptors in fish (Ngai et al., 1993); some 500 chemoreceptors in *C. elegans* (Troemel et al., 1995); a family of about 60 ORs in *Drosophila* and other insects (Clyne et al., 1999; Gao and Chess, 1999; Voss-hall et al., 1999); and the discovery of two families of GPCR genes for sweet, bitter, and amino acid ligands in the mammalian gustatory system (Hoon et al., 1999; Montmayeur et al., 2001; see Mombaerts, 2004, for an especially thorough review). All of these are GPCRs, some of the class A rhodopsin-like family, and some in the class C glutamate-type of receptor. None of them are otherwise closely related to the mammalian ORs. More recently, with the appearance of sequenced whole genomes for other species, data mining efforts have uncovered similarly large families of ORs in dog, chimpanzee, rat, and fugu. There are now several thousand GPCR genes encoding receptors for taste and smell across the fauna (Mombaerts, 2004).

The one area of promise not yet fully realized is that of understanding ligand selectivity in this large receptor family. At first, it seemed that the 1000 new GPCRs would produce novel insights in pharmacology. For years, the method of choice to understand GPCR structure-function was point mutagenesis, changing an amino acid and observing the effects on ligand binding. This strategy had met with some success in  $\beta$ 2-adrenergic receptors (Strader et al., 1987) and among the cone opsins (Nathans, 1990). But there are limits to this technique when it becomes necessary to make more than a single or double mutation at a time. Here was a family of receptors, some of which differed by only a few amino acids (greater than 98% identity) and others of which shared no more than 40% identity. It was as if evolution had run its own mutation experiment and selected the ones that worked. Taking advantage of the availability of huge chemical libraries, a high-throughput assay using receptors expressed in a heterologous cell system and challenged with thousands of odors would give a rapid readout of specificity that could be aligned with gene sequence. Matching odorants to their cognate receptors, and vice versa, would not only solve many of the mysteries of olfactory coding but also provide a deeper understanding of how certain residues interacted with ligands to determine specificity—an experimental program that could revolutionize the pharmacology of GPCRs. It should be noted in this regard that more than 50% of drugs used for medical purposes target GPCRs.

However, the catch to cashing in on all this bounty was that odorant receptors could not be expressed in heterologous cells. To be precise, OR protein is trapped in the endoplasmic reticulum and fails to traffic to the membrane, leaving the receptors nonfunctional. In spite of nearly 14 years of effort by many laboratories, the puzzle remains unsolved and the biology of the trafficking problem is still mysterious. To date, only a hand-

ful of ORs have been expressed in systems that allow their cognate ligands to be determined, and this with considerable difficulty (Touhara et al., 1999; Zhao et al., 1998). Some hope has been provided with the recent publication in *Cell* of a paper from Hiro Matsunami's lab (a former post doc with Buck) identifying two potential cofactors that appear to significantly enhance membrane expression for many ORs (Saito et al., 2004). But the kind of robust expression that can be achieved with many other GPCRs is not yet possible in ORs.

In neuroscience, sensory systems have played a key role in establishing fundamental principles of the nervous system. Concepts and systems such as receptive fields, labeled lines, lateral inhibition, topographical mapping, and even G protein-mediated transduction are a few of the seminal advances in understanding brain function that have come from the study of sensory systems, where the input is a physically known and measurable quantity and the neural output is often accessible for observation. Virtually all we know about the world comes to us through the holes in our heads (and the sheet of skin covering us). How the brain transforms physical stimuli into perception is at once one of the most profitable areas of investigation in neuroscience and one of its great remaining mysteries. Now that the olfactory system has been provided with a molecular basis for understanding detection and discrimination, it is possible to profitably investigate the wider questions of brain processing—is there a “chemical map” in the brain, how is intensity (concentration) separated from quality, how does the brain use neural space to encode an inherently nonspatial stimulus?

In his brief address at the Nobel banquet, the Physics prize winner David Gross astutely observed that “the most important product of science is ignorance.” This echoes a statement by an earlier laureate, Marie Curie, in a letter to her brother in 1894, that “one never notices what has been done; one can only see what remains to be done.” The work of Buck and Axel deserves its place in the upper echelons of the scientific pantheon, not only for what it discovered, but for the bountiful ignorance it has spawned, and for how it has allowed us to see all the work that needs to be done.

Two political postscripts. Linda Buck is only the seventh woman to be awarded the Nobel prize in Physiology or Medicine, a point that was underscored by Professor Bengt Samuelson in his opening address at the Nobel award ceremony. Samuelson contends that Nobel prizes reflect the academic and social realities of decades past and that the paucity of female laureates is a direct reflection of the difficulties women faced in pursuing academic and intellectual callings in the middle part of the 20th century. Along with Linda Buck, this year's prizes in Peace and in Literature were also awarded to women, perhaps, one hopes, presaging a change in cultural attitudes that will see more women in the academy and among Nobel's laureates.

Finally, Richard Axel's address at the Nobel banquet contains several noteworthy thoughts on the relation between ethics, morality, and free scientific inquiry in biomedical research. It is well worth reading, or watching, at the Nobel prize website (<http://nobelprize.org/>).

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